Neural Document Understanding: A Two-Column Study

Abstract

This paper presents an evaluation of layout analysis and text reconstruction techniques for complex document architectures. We demonstrate that column separation and reading order preservation are essential for accurate downstream processing.

1. Introduction

Modern document understanding pipelines face significant challenges when transcribing multi-column layouts. Without proper spatial reasoning, reading order across columns can easily become corrupted.

When reading a two-column page, a human reader naturally proceeds from the top of the left column to the bottom, before continuing to the top of the right column.

2. Methodology

Our approach groups tokens into lines and uses recursive XY-cut partitioning to identify column gutters and margins.

This ensures that independent columns are transcribed sequentially rather than interleaved horizontally.

3. Experimental Setup

We evaluated the pipeline across a diverse set of synthetic and real-world benchmark documents containing varied fonts and densities.